Scioto Community College

Chemistry | North Campus, Morse Rd, Columbus, OH | Department line (614) 555-0166

CHEM 1211 General Chemistry I Laboratory, Fall 2026

Tuesday 6:00 pm section (L03), Science Hall S-118 | Instructor: Paul Hendricks, phendricks@sciotocc.edu

Lab Report Grade Sheet

Lab 2: Calibration of a 50 mL Burette

SCORE
17 / 20

Student	Tomasz Nowak	**Performed**	Tue 09/29/2026
Bench partner	Mei Chen	**Submitted**	Tue 10/06/2026, start of lab
Section	L03, Tue 6:00 pm, S-118	**Graded**	Tue 10/13/2026
Instructor	Paul Hendricks	**Data sheet copy**	Attached

Rubric (lab report, 20 points)

Criterion	Possible	Earned	Notes
Data and tables	5	**3**	Table 2 volume read column reported to 0.1 mL (-1). Trial 3 final reading typed as 30.04 mL. Data sheet: 30.40 mL (-1).
Calculations	5	**5**	True volume from mass and density for each trial. Correction with correct sign. Sample calculation shown with units.
Results and discussion	5	**4**	Mean correction and direction stated correctly. “Human error” given as the error source (-1).
Format and prelab	5	**5**	Prelab complete. Procedure in past tense. Tables numbered, units in column headers.
Total	**20**	**17**	

Instructor comments

Table 2: report volumes to 0.01 mL. A burette reads to 0.01.

Trial 3 final reading: data sheet 30.40, Table 2 says 30.04. Check transcription, data sheet → report.

Read bottom of meniscus at eye level.

Discussion: “human error” is not an error source. Name one. Parallax, air bubble in tip, hanging drop.

Calculations clean.

-PH

Lab reports are due at the start of the next week's lab. Bench partners may share raw data. Calculations, tables and discussion are individual work.

Graded copy posted to Scioto Learn 10/13/2026. Marked-up report follows on pages 2 and 3.

Lab 2: Calibration of a 50 mL Burette

17/20

Format and prelab 5/5

Tomasz Nowak
CHEM 1211 Lab, Tuesday 6:00 pm, S-118 (P. Hendricks)
Bench partner: Mei Chen
Performed September 29, 2026

Purpose

To check the accuracy of a 50 mL burette by weighing the water it delivers and comparing the true volume with the volume read from the burette.

Introduction

The burette scale is marked every 0.1 mL, but the volume it actually delivers can differ slightly from the volume read off the scale. In this lab, water delivered from the burette was weighed, and its true volume was found by dividing the mass by the density of water at the measured temperature. The difference between the true volume and the volume read is the correction for that section of the burette.

Procedure

1. The 50 mL burette was rinsed with deionized water, filled above the 0.00 mL mark, and drained until the tip was free of air bubbles.
2. The temperature of the deionized water was measured as 23.0 °C, and its density, 0.99754 g/mL, was taken from the density table in the lab manual.
3. A clean, dry flask with a stopper was weighed on the balance, and the initial burette reading was recorded.
4. About 10 mL of water was delivered into the flask. The final reading was recorded, and the stoppered flask was weighed again.
5. Without refilling the burette, steps 3 and 4 were repeated for the next two 10 mL sections, for three trials total.

Read bottom of meniscus at eye level.

Data

3/5

Table 1. Water temperature and density

Measurement	Value
Temperature of deionized water	23.0 °C
Density of water at 23.0 °C (lab manual)	0.99754 g/mL

Table 2. Burette readings and mass of water delivered

Trial	Initial reading (mL)	Final reading (mL)	Volume read (mL)	Mass of water (g)	True volume (mL)	Correction (mL)
1	0.20	10.25	10.1	10.01	10.03	-0.02
2	10.25	20.31	10.1	10.03	10.05	-0.01
3	20.31	30.04	10.1	10.05	10.07	-0.02

Report to 0.01 mL. A burette reads to 0.01.

Data sheet: 30.40. Check transcription.

Sample calculation (Trial 1) **5/5**

Volume read = final reading - initial reading
= 10.25 mL - 0.20 mL = 10.05 mL ✓

True volume = mass of water / density
= 10.01 g / 0.99754 g/mL = 10.03 mL ✓

Correction = true volume - volume read
= 10.03 mL - 10.05 mL = -0.02 mL ✓

Results

Corrections for the three trials were -0.02 mL, -0.01 mL and -0.02 mL. Mean correction for a 10 mL delivery: -0.02 mL. The burette delivers slightly less than it reads. ✓

Discussion **4/5**

All three corrections are small and in the same direction, so the burette is accurate to about 0.02 mL over 10 mL. Over the three trials, 30.20 mL was read and 30.15 mL was delivered, a total correction of -0.05 mL from 0 to 30 mL. The differences between trials are most likely human error in reading the burette.

Not an error source. Name one: parallax, air bubble in tip, hanging drop.

17/20. See grade sheet. -PH